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Safety-Enhanced UAV Path Planning Using Spherical Vector-Based Particle Swarm Optimization

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Stochastic UAV Path Planning using SPSO under Uncertainty

Objective
→ Find optimal UAV path from start to destination

Constraints
→ Obstacle avoidance
→ Smooth path
→ Communication quality

Application
→ Drone delivery
→ Surveillance
→ Disaster response

Real-world environments are uncertain thus needs randomness

Deterministic Baseline (M3)

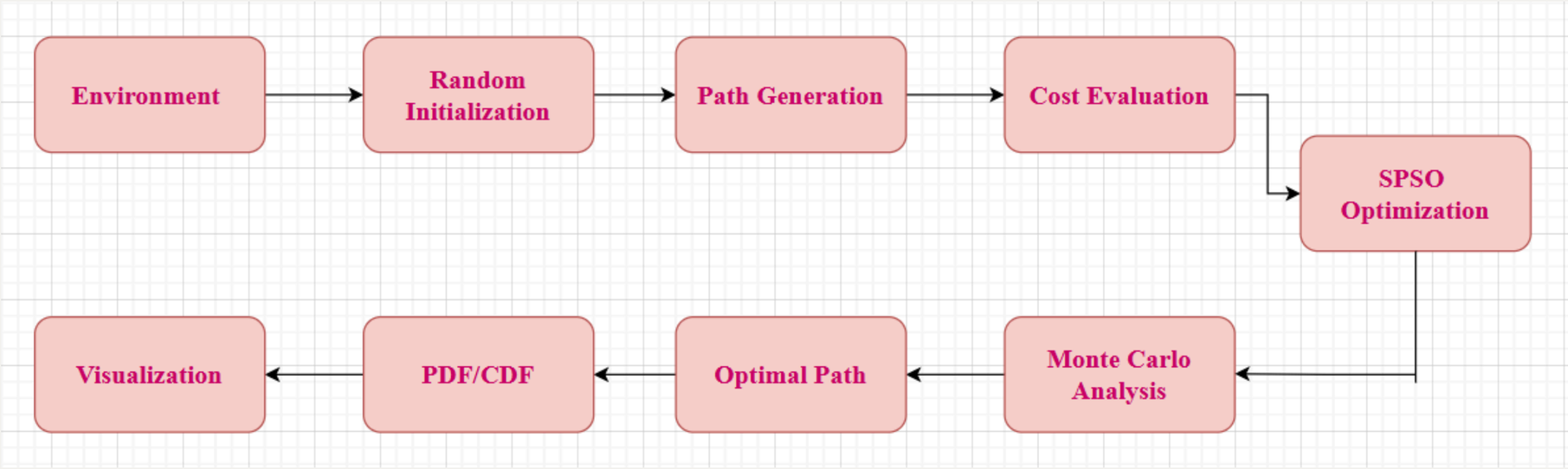
Content

→ Used SPSO (Swarm Optimization)
→ Fixed obstacles
→ Perfect UAV position
→ No noise or uncertainty
→ Single-run evaluation

Limitation

→ Not realistic
→ Cannot handle dynamic environments
→ Over-optimistic performance

System Pipeline



- Path generation (particles)
- Cost function evaluation
- Optimization loop
- Final path visualization

Decision happens at COST FUNCTION

Sources of Uncertainty

Content

→ Moving obstacles (birds, kites, planes)
→ Wind disturbance
→ UAV position noise
→ Communication uncertainty (LoS/NLoS)
→ Random initialization

Categorize

→ Environmental
→ Measurement
→ Communication

Random Variables Probabilistic Modeling

Component	Random Variable
UAV motion	Gaussian noise
Wind	Random vector
Obstacles	Random movement
Communication	Probabilistic LoS
SNR	Gaussian noise

Distributions used:

- **Gaussian → randn()**
- **Uniform → rand()**

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Randomized Algorithm Design

- PSO with stochastic updates

→ **Random velocity update**

$$v = wv + r1(pbest - x) + r2(gbest - x)$$

→ **Noise added to path:**

$$\text{path} + \text{Gaussian Noise} + \text{wind}$$

- **Moving obstacles are updated randomly**

Probabilistic Reasoning

- Because of Randomization the UAV explored multiple possible paths
- Prevents SPSO from getting stuck in local minima
- Improves robustness against:
  - Moving obstacles
  - Wind disturbance
  - Noisy UAV position
- Makes the planner adaptable to dynamic environments

Mathematical Strategy

$$v_{t+1} = wv_t + r_1(pbest - x_t) + r_2(gbest - x_t)]$$

Random coefficients improve exploration + exploitation
Expected cost decreases over iterations:

$$E[C_{t+1}] \leq E[C_t]$$

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Implementation (Code Highlights)

Monte Carlo + Random Initialization

```
for run = 1:runs
    particles = rand(num_particles, num_waypoints, 3)*area_size;
```

- Repeats the experiment using a number of randomized runs.
- Prepares various UAV routes to be analyzed by Monte Carlo.

Dynamic Obstacles + Noise

```
birds = birds + randn(size(birds));
wind = wind_strength * randn(size(path));
noisy_path = path + sigma*randn(size(path)) + wind;
```

- Birds and obstacles are moving in a haphazard manner.
- UAV uncertainty, wind, Gaussian noise.

SPSO Velocity Update

```
velocities = 0.5*v + r1*(pbest - path) + r2*(gbest - path);
```

- Core SPSO update rule
- Random coefficients enhance an exploration of paths.

Implementation (Code Highlights)

Multi-Objective Cost Function

```
cost = cost + norm(path(i+1,:) - path(i,:));
if norm(path(j,:) - obstacles(i,:)) < 5
    cost = cost + 100;
end
```

- Minimizes path length
- Applies penalty on heavy collision when avoiding obstacles.

LoS + SNR Communication Model

```
P_LoS = 1/(1 + exp(-0.1*(theta - 0.5)));
SNR = Pt*G/N0 + randn()*0.5;
```

- Probabilistic LoS/NLoS link
- Random SNR variation of realistic communication.

Evaluation Metrics

Path & Safety Metrics

- Total Cost (C) → Overall performance
- Path Length (J_1) → Efficiency
- Minimum Obstacle Distance → Safety

Communication Metrics

- Average SNR (γ) → Signal Quality
- Data Rate (R) → Communication Performance

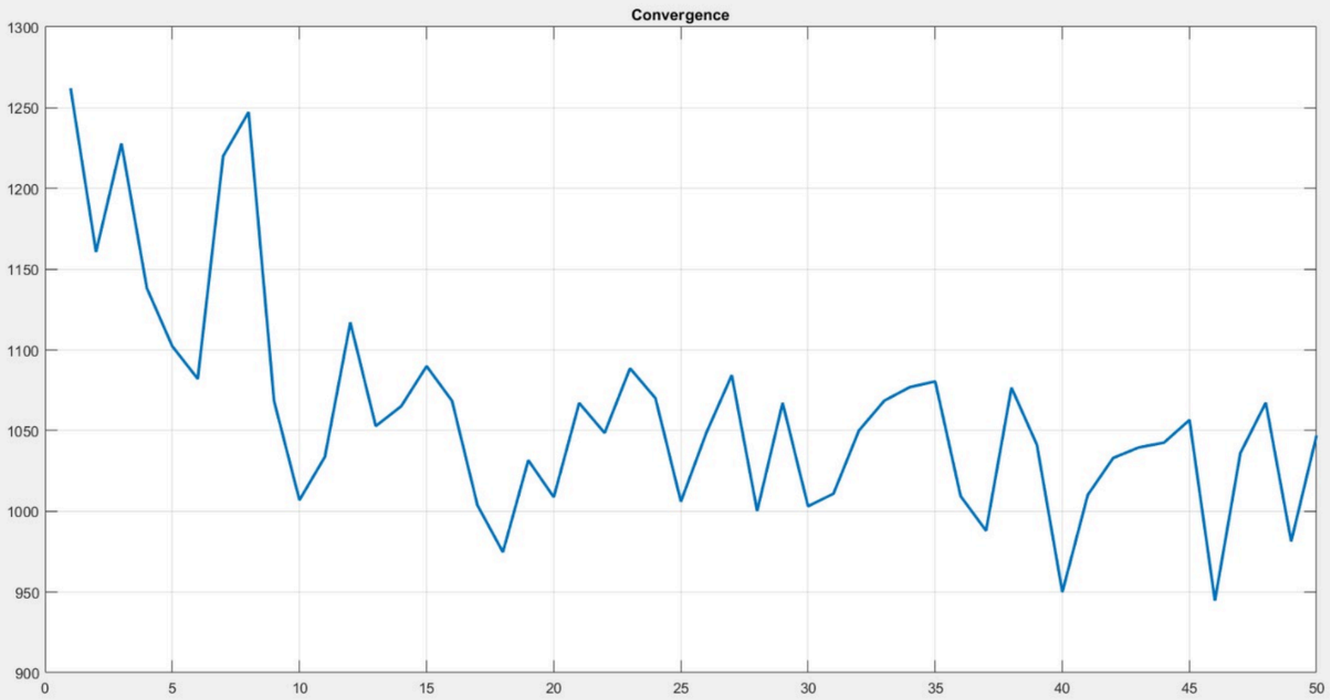
Optimization & Analysis

- Convergence Curve → Optimization Behavior
- PDF → Distribution of Results
- CDF → Probability Analysis

Statistical Metrics

- Mean Cost → Efficiency
- Standard Deviation → Variability
- Collision Rate → Safety Reliability

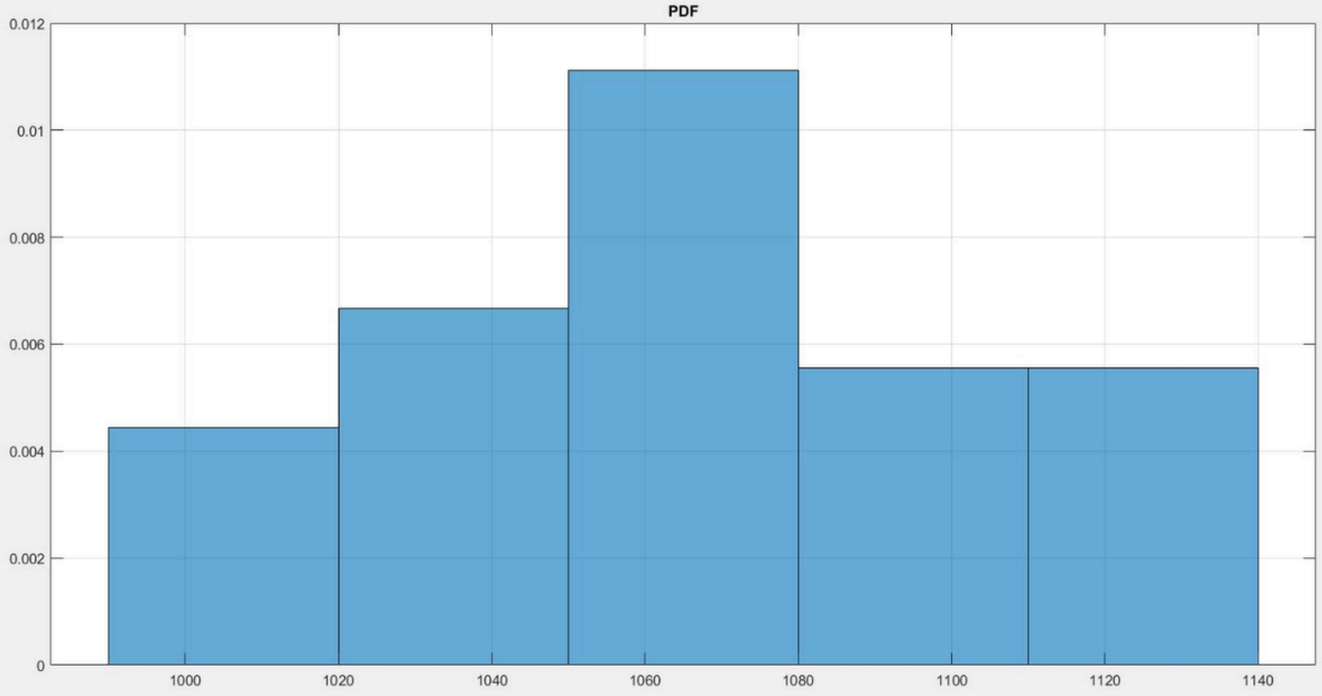
Convergence



Interpretation:

→ The cost <u>decreases rapidly</u> during initial iterations (exploration phase)
→ The <u>solution stabilizes</u> in later iterations (convergence phase)
→ <u>Small fluctuations persist</u> due to stochastic components in the algorithm

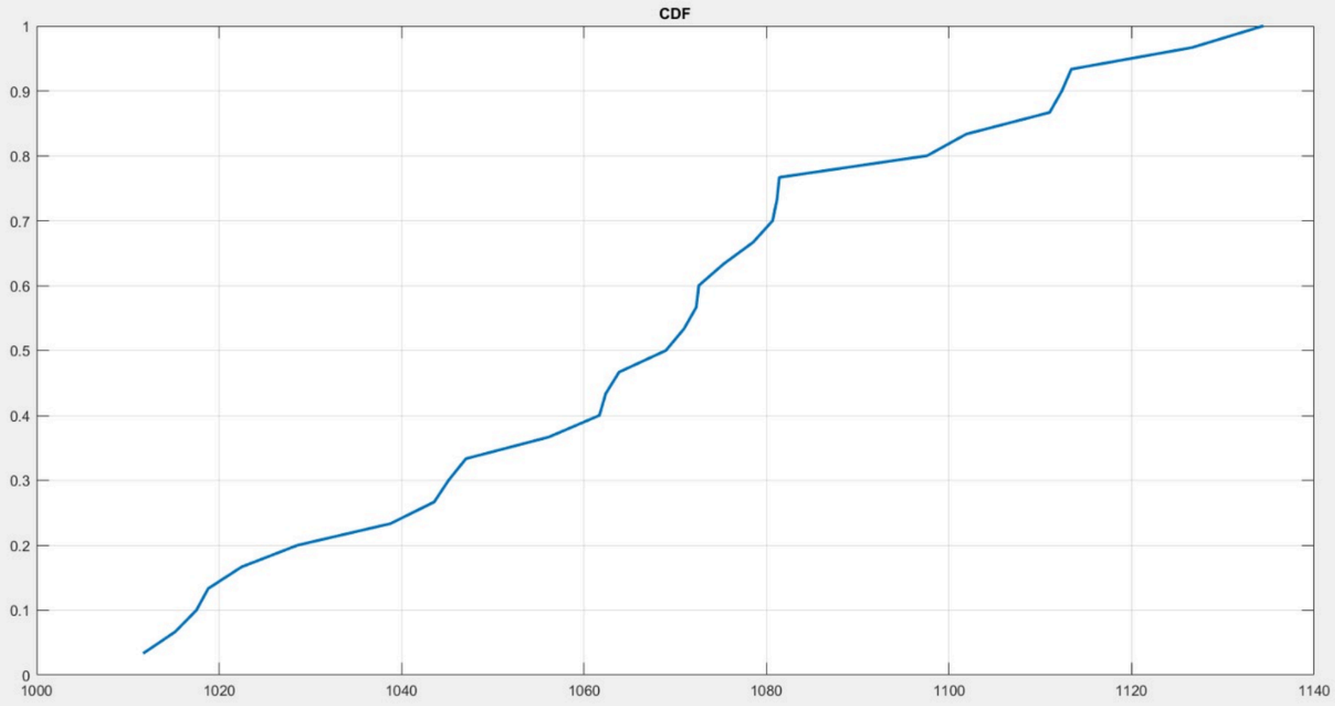
Probability Density Function



Interpretation:

→ Distribution is centered around optimal cost region
→ Narrow spread indicates low variability
→ Consistent results across multiple runs
→ Demonstrates robustness under randomness

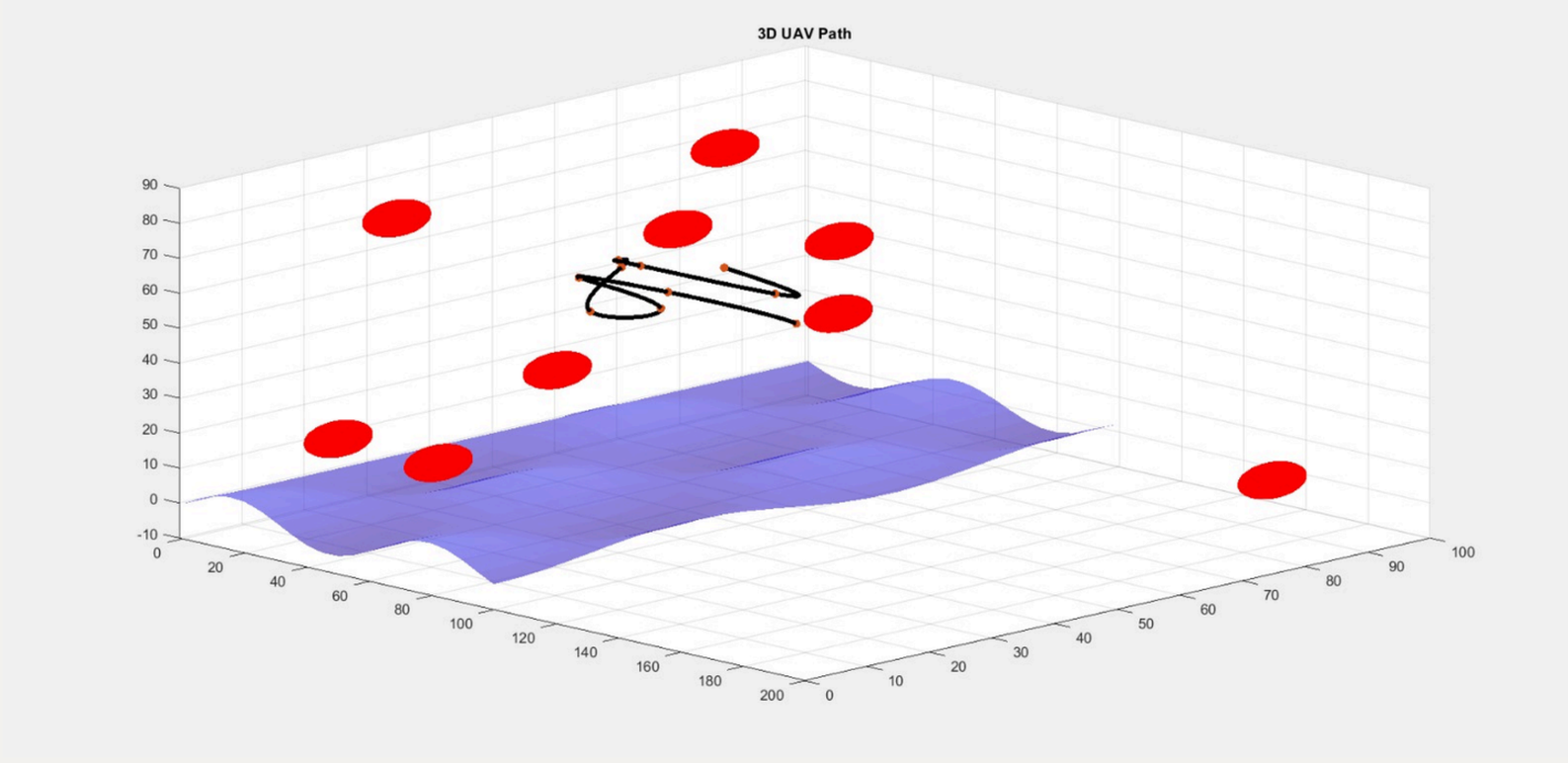
Cumulative Distribution Function



Interpretation:

→ The CDF shows the probability of achieving cost below a given value.
→ The curve increases smoothly, indicating stable and consistent performance.
→ The majority of solutions lie within a narrow cost range (~1050–1100).
→ High probability of near-optimal solutions with low likelihood of poor outcomes.

3D Visualization



Content:

→ Path avoids all obstacles → collision-free
→ Smooth trajectory → feasible UAV motion
→ Maintains safe altitude above terrain
→ The optimization effectively balances safety, smoothness, and feasibility.

Comparison: M3 VS M4

Milestone 3	Milestone 4
Ideal environment	Realistic environment with uncertainty
Cost Function: $C = b_1 J_1 + b_2 J_2 + b_3 J_3 + b_4 J_4$	Cost Function: $C_{new} = C + \lambda \cdot \frac{1}{R}$
Stattionary UAV	Dynamic UAV
Single-run simulation	Monte Carlo simulation
standard SPSO (deterministic cost evaluation)	Stochastic SPSO (randomized inputs + multiple runs)

Key Insights

Performance

- Consistent performance under uncertainty
- Low variance → stable results
- Smooth and efficient path optimization

Safety

- Zero collision rate across simulations
- Maintains safe clearance from obstacles
- Respects turning and climbing constraints

Robustness

- Randomness improves adaptability
- Effective exploration avoids local minima
- Stable convergence behavior

Validation & Analysis

- Monte Carlo ensures reliable evaluation
- PDF indicates low variability
- CDF confirms high probability of near-optimal solutions

Thank you!!